

大语言模型部署体验与横向对比实验报告

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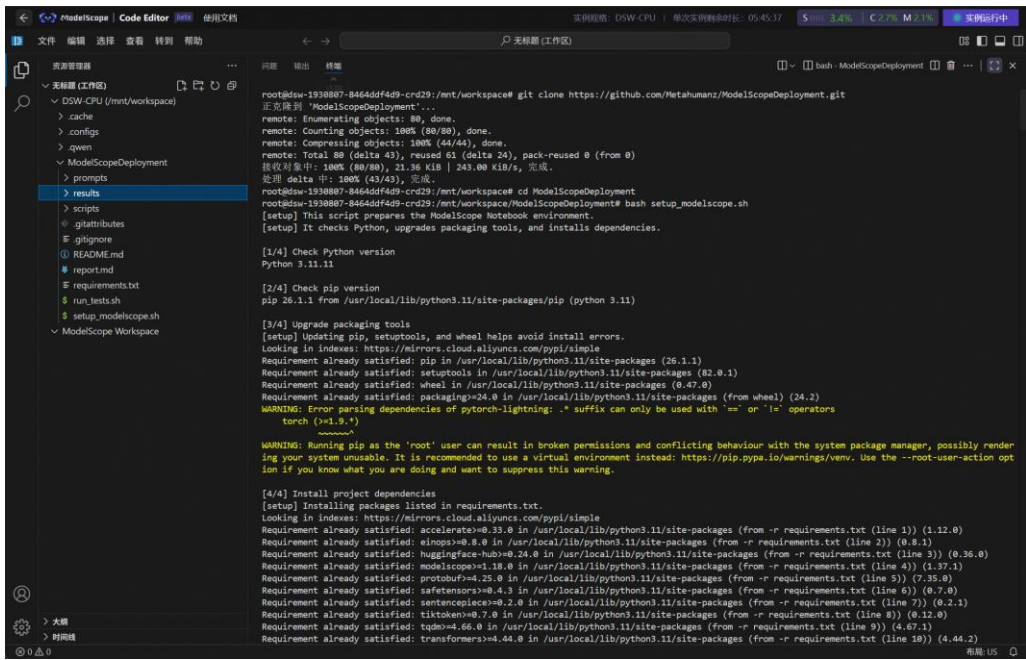
一、项目公开链接与实验目的

公开仓库地址：<https://github.com/Metahumanz/ModelScopeDeployment>

本实验在 ModelScope Notebook CPU 环境中完成模型部署、依赖安装和中文语义理解问答测试。结合免费 CPU 实例的运行时长限制，本次实验采用 0.5B 到 1.5B 的轻量模型进行横向比较。

二、实验环境与部署过程

项目	内容
实验平台	ModelScope Notebook
推荐镜像	ubuntu22.04-py311-torch2.3.1-1.37.1
运行方式	CPU 推理
主要依赖	modelscope、transformers、torch、accelerate、sentencepiece 等
统一入口	bash setup_modelscope.sh; bash run_tests.sh



```
root@dsw-193887-8464ddf4d9-crd29:/mnt/workspace# git clone https://github.com/Metahumanz/ModelScopeDeployment.git
正在检出 'ModelScopeDeployment'...
remote: Enumerating objects: 88, done.
remote: Counting objects: 100% (88/88), done.
remote: Compressing objects: 100% (44/44), done.
remote: Total 88 (delta 43), reused 61 (delta 24), pack-reused 0 (from 0)
接收对象中: 100% (88/88), 21.36 KiB | 243.00 KiB/s, 完成。
处理 delta 中: 100% (43/43), 完成。
root@dsw-193887-8464ddf4d9-crd29:/mnt/workspace# cd ModelScopeDeployment
root@dsw-193887-8464ddf4d9-crd29:/mnt/workspace/ModelScopeDeployment# bash setup_modelscope.sh
[setup] This script prepares the ModelScope Notebook environment.
[setup] It checks Python, upgrades packaging tools, and installs dependencies.

[1/4] Check Python version
Python 3.11.11

[2/4] Check pip version
pip 26.1.1 from /usr/local/lib/python3.11/site-packages/pip (python 3.11)

[3/4] Upgrade packaging tools
[setup] Updating pip, setuptools, and wheel helps avoid install errors.
Looking in indexes: https://mirrors.cloud.aliyuncs.com/pypi/simple
Requirement already satisfied: pip in /usr/local/lib/python3.11/site-packages (26.1.1)
Requirement already satisfied: setuptools in /usr/local/lib/python3.11/site-packages (82.0.1)
Requirement already satisfied: wheel in /usr/local/lib/python3.11/site-packages (0.47.0)
Requirement already satisfied: packaging>24.0 in /usr/local/lib/python3.11/site-packages (from wheel) (24.2)
WARNING: Error parsing dependencies of pytorch-lightning: .* suffix can only be used with == or != operators
torch (>=1.9.*)
~~~~~
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager, possibly render
ing your system unstable. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv. Use the --root-user-action opt
ion if you know what you are doing and want to suppress this warning.

[4/4] Install project dependencies
[setup] Installing packages listed in requirements.txt.
Looking in indexes: https://mirrors.cloud.aliyuncs.com/pypi/simple
Requirement already satisfied: accelerate>0.33.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 1)) (1.12.0)
Requirement already satisfied: einops>0.8.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 2)) (0.8.1)
Requirement already satisfied: huggingface-hub>0.24.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 3)) (0.36.0)
Requirement already satisfied: modelscope>1.18.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 4)) (1.37.1)
Requirement already satisfied: protobuf>4.25.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 5)) (7.35.0)
Requirement already satisfied: safetensors>0.4.3 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 6)) (0.7.0)
Requirement already satisfied: sentencepiece>0.2.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 7)) (0.2.1)
Requirement already satisfied: tiktoken>0.7.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 8)) (0.12.0)
Requirement already satisfied: tqdm>4.66.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 9)) (4.67.1)
Requirement already satisfied: transformers>4.44.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 10)) (4.44.2)
```

图1 ModelScope Notebook 工作界面

```
root@baw-1586807-8646f4bf-c029:/mnt/norwspace# git clone https://github.com/Netahuman/ModelScopeDeployment.git
正在克隆 'ModelScopeDeployment'...
remote: Enumerating objects: 80, done.
remote: Counting objects: 100% (80/80), done.
remote: Compressing objects: 100% (46/46), done.
remote: Total 80 (delta 40), reused 0 (delta 0), pack-reused 0 (from 0)
收到 17 个文件: 100.0% (delta 0), 完成
收到 delta #1: 100% (41/41), 完成
remote: 1586807-8646f4bf-c029:/mnt/norwspace/ModelScopeDeployment# bash setup_modelscope.sh
[info] This script prepares the ModelScope deployment environment.
[info] It checks Python, upgrades packaging tools, and installs dependencies.

[1/4] Check Python version
Python 3.11.11

[2/4] Check pip version
pip 23.1.1 from /usr/local/lib/python3.11/site-packages/pip (python 3.11)

[3/4] Upgrade packaging tools
[warning] Updating pip, setuptools, and wheel helps avoid install errors.
Linking in indexes: https://mirrors.cloud.aliyuncs.com/pypi/simple
Requirement already satisfied: pip in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 1)) (23.1.1)
Requirement already satisfied: setuptools in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 2)) (68.1.1)
Requirement already satisfied: wheel in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 3)) (0.40.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 4)) (24.1)
WARNING: Error parsing dependencies of pytorch-lightning: .* suffix can only be used with == or >= operators
torch (2.0.1)

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager, possibly rendering your system unusable. It is recommended to use a virtual environment instead: https://pip.pypa.io/en/latest/using/#use-the-root-user-action-apt
but if you know what you are doing and want to suppress this warning.

[4/4] Install project dependencies
[info] Installing packages listed in requirements.txt
Linking in indexes: https://mirrors.cloud.aliyuncs.com/pypi/simple
Requirement already satisfied: accelerate>=0.18.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 1)) (0.12.0)
Requirement already satisfied: einops>=0.6 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 2)) (0.6.1)
Requirement already satisfied: huggingface-hub>=0.6 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 3)) (0.36.0)
Requirement already satisfied: modelscope>=1.18.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 4)) (1.19.1)
Requirement already satisfied: protobuf>=3.9 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 5)) (3.20.0)
Requirement already satisfied: safetensors>=0.4 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 6)) (0.2.9)
Requirement already satisfied: sentencepiece>=0.1.9 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 7)) (0.1.9)
Requirement already satisfied: tiktoken>=0.7.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 8)) (0.12.0)
Requirement already satisfied: transformers>=4.26.0 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 9)) (4.42.2)
Requirement already satisfied: transformers<=4.40 in /usr/local/lib/python3.11/site-packages (from -r requirements.txt (line 10)) (4.44.1)
Requirement already satisfied: numpy<=1.17 in /usr/local/lib/python3.11/site-packages (from accelerate>=0.18.0->-r requirements.txt (line 1)) (1.26.4)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/site-packages (from accelerate>=0.18.0->-r requirements.txt (line 1)) (24.1)
Requirement already satisfied: psutil in /usr/local/lib/python3.11/site-packages (from accelerate>=0.18.0->-r requirements.txt (line 1)) (7.0.0)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.11/site-packages (from accelerate>=0.18.0->-r requirements.txt (line 1)) (6.0.2)
Requirement already satisfied: torch>=2.0 in /usr/local/lib/python3.11/site-packages (from accelerate>=0.18.0->-r requirements.txt (line 1)) (2.0.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/site-packages (from huggingface-hub>=0.6->-r requirements.txt (line 3)) (3.12.0)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/site-packages (from huggingface-hub>=0.6->-r requirements.txt (line 3)) (2023.9.0)
Requirement already satisfied: requests in /usr/local/lib/python3.11/site-packages (from huggingface-hub>=0.6->-r requirements.txt (line 3)) (2.31.0)
Requirement already satisfied: typing-extensions>=4.3 in /usr/local/lib/python3.11/site-packages (from huggingface-hub>=0.6->-r requirements.txt (line 3)) (4.12.2)
Requirement already satisfied: hf-vc2-0.6, <=1.5 in /usr/local/lib/python3.11/site-packages (from huggingface-hub>=0.6->-r requirements.txt (line 3)) (1.2.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.11/site-packages (from modelscope>=1.18.0->-r requirements.txt (line 4)) (68.1.1)
Requirement already satisfied: urllib3>=1.26 in /usr/local/lib/python3.11/site-packages (from modelscope>=1.18.0->-r requirements.txt (line 4)) (2.0.7)
Requirement already satisfied: requests>=2021.1.18 in /usr/local/lib/python3.11/site-packages (from tiktoken>=0.7.0->-r requirements.txt (line 8)) (2023.11.4)
Requirement already satisfied: sentencepiece>=0.1.9 in /usr/local/lib/python3.11/site-packages (from transformers>=4.26.0->-r requirements.txt (line 9)) (0.1.9)
Requirement already satisfied: charset-normalizer<=3 in /usr/local/lib/python3.11/site-packages (from requests>=2021.1.18->-r requirements.txt (line 3)) (3.4.1)
Requirement already satisfied: idna<=3.5 in /usr/local/lib/python3.11/site-packages (from requests>=2021.1.18->-r requirements.txt (line 3)) (3.10)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/site-packages (from requests>=2021.1.18->-r requirements.txt (line 3)) (2023.11.17)
Requirement already satisfied: sympy in /usr/local/lib/python3.11/site-packages (from torch>=2.0.0->accelerate>=0.18.0->-r requirements.txt (line 1)) (1.12.0)
Requirement already satisfied: networkx in /usr/local/lib/python3.11/site-packages (from torch>=2.0.0->accelerate>=0.18.0->-r requirements.txt (line 1)) (3.2.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.11/site-packages (from torch>=2.0.0->accelerate>=0.18.0->-r requirements.txt (line 1)) (3.1.3)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/site-packages (from Jinja2>=2.0.0->accelerate>=0.18.0->-r requirements.txt (line 1)) (2.1.2)
Requirement already satisfied: qmshash<1.4, >=1.1.0 in /usr/local/lib/python3.11/site-packages (from sympy>=1.12.0->-r requirements.txt (line 1)) (1.3.0)
WARNING: Error parsing dependencies of pytorch-lightning: .* suffix can only be used with == or >= operators
torch (2.0.1)

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager, possibly rendering your system unusable. It is recommended to use a virtual environment instead: https://pip.pypa.io/en/latest/using/#use-the-root-user-action-apt
but if you know what you are doing and want to suppress this warning.

[info] Installed runtime versions:
torch: 2.0.1
transformers: 4.42.2
modelscope: 1.19.1

[info] Done. Next step:
bash run_tests.sh
root@baw-1586807-8646f4bf-c029:/mnt/norwspace/ModelScopeDeployment#
```

图2 克隆已上传的公开仓库并执行环境部署脚本

三、测试模型与问题集

标签	模型	参数规模	用途
qwen2.5-0.5b	qwen/Qwen2.5-0.5B-Instruct	0.5B	最新小模型，测试部署和基础中文理解
qwen2-0.5b	qwen/Qwen2-0.5B-Instruct	0.5B	Qwen2 代际对照
qwen1.5-0.5b	qwen/Qwen1.5-0.5B-Chat	0.5B	旧版小模型对照
qwen2.5-1.5b	qwen/Qwen2.5-1.5B-Instruct	1.5B	更大轻量模型，观察回答完整性提升

默认问题集位于 prompts/semantic_understanding.json，共 5 个问题，覆盖中文歧义理解、双关与语义反转、多层嵌套指代、人物指代关系和词义消歧。

编号	测试重点	问题摘要
1	中文歧义理解	请说出以下两句话区别在哪里？1、冬天：能穿多少穿多少。2、夏天：能穿多少穿多少。
2	双关与语义反转	请说出以下两句话区别在哪里？单身狗产生的原因有两个，一是谁都看不上，二是谁都看不上。
3	多层嵌套指代	他知道我知道你知道他不知道吗？这句话里，到底谁不知道？
4	人物指代关系	明明明明明白白白喜欢他，可她就是不说。这句话里，明明和白白谁喜欢谁？
5	词义消歧	领导：你这是什么意思？小明：没什么意思。意思意思。领导：你这就不够意思了。小明：小意思，小意思。领导：你这人真有意思。小明：其实也没有别的意思。领导：...

四、运行截图与结果保存

每个模型的完整输出保存到 results/<label>/results.md 和 results/<label>/results.json。以下截图来自根目录保存的实际运行画面。

Qwen2.5-0.5B-Instruct



图3 Qwen2.5-0.5B-Instruct 问答测试截图

Qwen2-0.5B-Instruct

```

[tests] Running qwen2-0.5b
[tests] Model: qwen/Qwen2-0.5B-Instruct
[tests] The model will answer every question, then save MarkDown and JSON results.
=====
[eval] Loading question set.
[eval] Loaded 3 questions from prompts/semantic_understanding.json.
[eval] Resolving model path. This may download the model if it is not cached.
[modelscope] Downloading or locating qwen/Qwen2-0.5B-Instruct
[modelscope] Downloading Model from https://www.modelscope.cn to directory: /mnt/workspace/.cache/modelscope/models/qwen/Qwen2-0.5B-Instruct
2026-08-27 15:43:17,136 - modelscope - INFO - Got 10 files, start to download ...
Downloading [README.md] 100% | 3.40k/3.40k [00:00:00.00, 11.50k/s]
Downloading [LICENSE] 100% | 11.1k/11.1k [00:00:00.00, 35.20k/s]
Downloading [configuration.json] 100% | 48.0/48.0 [00:00:00.00, 1410k/s]
Downloading [config.json] 100% | 650/650 [00:00:00.00, 1.520k/s]
Downloading [merges.txt] 100% | 1.59M/1.59M [00:00:00.00, 4.49M/s]
Downloading [tokenizer.json] 100% | 6.70M/6.70M [00:00:00.00, 13.76M/s]
Downloading [tokenizer_config.json] 100% | 1.20k/1.20k [00:00:00.00, 4.80k/s]
Downloading [generation_config.json] 100% | 242/242 [00:00:00.00, 330M/s]
Downloading [vocab.json] 100% | 2.65M/2.65M [00:00:00.00, 6.97M/s]
Downloading [model.safetensors] 100% | 942M/942M [00:06:00.00, 140M/s]
Processing 10 items: 100% | 10.0/10.0 [00:00:00.00, 1.40k/s]
2026-08-27 15:43:28,116 - modelscope - INFO - Finish downloading 10 files for repo 'qwen/Qwen2-0.5B-Instruct'
2026-08-27 15:43:28,118 - modelscope - INFO - Creating symbolic link [/mnt/workspace/.cache/modelscope/models/qwen/Qwen2-0.5B-Instruct]., 1.68M/s]
[eval] Loading tokenizer and model into CPU memory with torch dtype: float32, torch threads: 4. | 10.0M/942M [00:00:00.52, 18.0M/s]
[load] torch: 2.3 items | 933M/942M [00:06:00.00, 150M/s]
[load] transformers: 4.64.20rs: | 99%
[load] torch threads: 4
[load] tokenizer: /mnt/workspace/.cache/modelscope/models/qwen/Qwen2-0.5B-Instruct
[load] model: /mnt/workspace/.cache/modelscope/models/qwen/Qwen2-0.5B-Instruct
[eval] Model is ready. Starting question answering.

=====
[1/3] 中文歧义理解
请选出以下两句话区别在哪里？1. 冬天，能穿多少穿多少。2. 夏天，能穿多少穿多少。
-----
[eval] Generating answer with max_new_tokens=1024.
[answer] 这两句都是关于穿衣的，但它们表达的意思不同。
1. “冬天”是说冬天可以穿很多衣服，而“夏天”则是说夏天可以穿很多衣服。
2. “冬天”是说冬天可以穿很多衣服，“夏天”则是在强调夏天也可以穿很多衣服。
[eval] Generation finished in 4.1s.

=====
[2/3] 褒义与贬义反转
请选出以下两句话区别在哪里？单身狗产生的原因有两个，一是谁都看不上，二是谁都看不上。
-----
[eval] Generating answer with max_new_tokens=1024.
[answer] 这句话的语义是：单身狗产生的原因有两个，一个是大家都看不上他（她），另一个是因为大家都看不好他（她）。
[eval] Generation finished in 1.7s.

=====
[3/3] 多轮对话指代
他知道我知道他知道他不知道吗？这句话，到底谁不知道？
-----
[eval] Generating answer with max_new_tokens=1024.
[answer] 这句话他知道他知道他不知道吗？“中”的“他知道”是主语，“我知道”是谓语，“他知道”是宾语，所以这句话的主语是“我”，谓语是“知道”，宾语是“他”。
[eval] Generation finished in 3.0s.

=====
[4/3] 人物指代关系
明明明明白白白白喜欢你，可她就是不谈。这句话里，明明和白白谁喜欢你？
-----
[eval] Generating answer with max_new_tokens=1024.
[answer] 这个句子的主语是“明明”，宾语是“白白”，所以，明明喜欢白白。
[eval] Generation finished in 1.4s.

=====
[5/3] 同义替换
领导：那这是什么意思？小明：没什么意思，就是套话。领导：你就不够意思了。小明：小意思，小意思。领导：你这个人真有意思。小明：其实也没有别的意思。领导：那我就不好意思了。小明：是就不好意思。请问：以上“意思”分别是什么意思？
-----
[eval] Generating answer with max_new_tokens=1024.
[answer] “意思”是表示同意或赞同“的意思”，“小意思”是指“不重要的事情”，“小意思”与“意思”相对应，“领导”是“上级”的意思，“领导”与“意思”相对应，“就不好意思”是“不好意思”的意思，“不好意思”与“意思”相对应。
[eval] Generation finished in 3.9s.
[eval] Writing result files.
=====
[eval] Saved MarkDown and JSON results to: results/qwen2-0.5b

```

图 4 Qwen2-0.5B-Instruct 问答测试截图

Qwen1.5-0.5B-Chat

[illegible]

图5 Qwen1.5-0.5B-Chat 问答测试截图

Qwen2.5-1.5B-Instruct

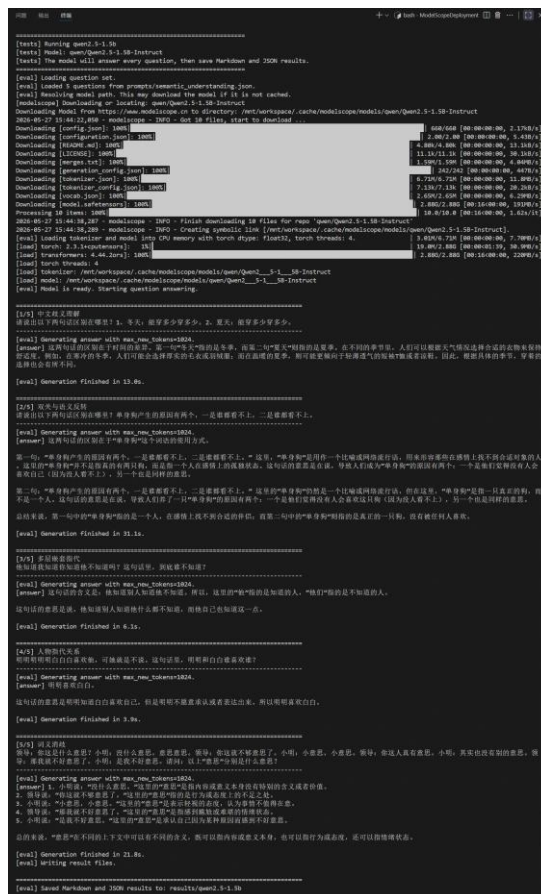


图6 Qwen2.5-1.5B-Instruct 问答测试截图

五、结果分析

从结果看，轻量模型能够完成基本格式化回答，但对中文语用双关、深层指代和多义词拆解的稳定性仍有限。相对而言，Qwen2.5-1.5B 的回答更完整，但也会出现解释方向偏差；0.5B 模型速度更适合 CPU 实验，但复杂语义能力不稳定。

维度	qwen1.5-0.5b	qwen2-0.5b	qwen2.5-0.5b	qwen2.5-1.5b
中文歧义理解	能区分季节，但未抓住语用反差	能给出简短解释，但较浅	偏字面解释季节差异	解释更完整，结合穿衣场景
双关与语义反转	理解偏差较明显	能接近“谁看不上谁”的双关	基本未解释核心双关	篇幅更完整，但出现字面化误读
多层嵌套指代	回答较模糊	语法分析错误	出现严重重复	能尝试解释但结论仍不稳

人物指代关系	误判为双方喜欢	判断为明明喜欢白白	判断为明明喜欢白白	判断为明明喜欢白白，但解释不够严谨
词义消歧	只能给出笼统解释	解释混乱	只概括大类含义	能列举多种“意思”的语境含义，表现最好
综合表现	旧版基线，稳定性一般	简洁但深度不足	部分问题出现重复或浅层解释	综合最好，但 CPU 成本更高

六、典型问题观察

- 0.5B 模型在简单问答中能输出可读中文，但对“能穿多少穿多少”这类需要结合常识和语用反差的问题，常停留在季节字面解释。
- 多层嵌套指代是最困难的问题，多个模型都没有稳定拆出“他不知道”的逻辑层级，Qwen2.5-0.5B 还出现明显重复。
- 词义消歧中，Qwen2.5-1.5B 的表现明显更好，能把“意思”拆成含义、态度、客气/不好意思等多个方向。
- 在 ModelScope 免费 CPU 环境中，轻量模型更适合作为课程实验对象，能够在有限时间内完成部署、问答和结果保存。

七、实验结论

1. 仓库已经提供可复现的部署脚本、统一测试脚本、问题集和结果保存目录，公开仓库地址为 <https://github.com/Metahumanz/ModelScopeDeployment>。
2. Qwen2.5-1.5B-Instruct 在回答完整性和词义消歧上相对最好，适合作为轻量模型对比中的主模型。
3. Qwen2.5-0.5B、Qwen2-0.5B 和 Qwen1.5-0.5B 更适合快速完成部署验证和截图，但复杂语义理解能力有限。
4. 免费 CPU Notebook 更适合运行轻量模型，本实验选择的模型能满足课程作业中部署体验、问答截图和横向比较的需求。